

PDF Report: Lab 1

The first lab of the semester presented a few difficulties throughout the process of completing it. Nevertheless, we still managed to complete the lab and full, hopefully meeting your requirements.

Going over task 1 first, the first thing of note was the amount of code that was given to us. It helped a lot in being able to just focus on the technical parts discussed in class. In our actual function “void *print_vowels(void *arg)” the first thing we did was add our mutex lock on the line that says “pthread_mutex_lock(&mutex). After which we then added the unlock to pair alongside it: “pthread_mutex_unlock”. This, of course, needed to be written directly after finishing the while loop, otherwise we’d enter an infinite loop.

```
PS C:\Users\Sabil_cljn5z9\OneDrive\Desktop\CSC_112_Projects> ./task1 charizard evee salamance axew kyurem
Consonant: charizard
Vowel: evee
Consonant: salamance
Vowel: axew
Consonant: kyurem
PS C:\Users\Sabil_cljn5z9\OneDrive\Desktop\CSC_112_Projects> ./task1 apple banana orange grape kiwi umbrella
Vowel: apple
Consonant: banana
Vowel: orange
Consonant: grape
Consonant: kiwi
Vowel: umbrella
```

With task 1 complete, the next focus for us was task 2. This one came with a lot less struggle than task 1 because once we “warmed up”, for a lack of a better term, we felt much more ready and confident. For task 2, understanding the concept of what needed to be done was certainly the “hardest part”. The solution to the conceptual problem was looping through an array of user Input words each of which would be checked if they were vowels or consonants.

```
PS C:\Users\Sabil_cljn5z9\OneDrive\Desktop\CSC_112_Projects> ./task2 apple banana orange grape kiwi umbrella
Vowel: apple
Consonant: banana
Consonant: grape
Consonant: kiwi
Vowel: orange
Vowel: umbrella
```

```
PS C:\Users\Sabil_cljn5z9\OneDrive\Desktop\CSC_112_Projects> ./task2 charizard evee salamance axew kyurem
Vowel: evee
Vowel: axew
Consonant: charizard
Consonant: salamance
Consonant: kyurem
```

They would then be able to successfully be printed in order of receipt but also selectively with vowels first and then only consonants.

Having completed two of the three tasks we turned our attention to the only one remaining. Task 3 was by far the most challenging for our group. This may not be the case for all groups but no one in our group had a solid enough foundation of child processes to make anything work. The main question we got stuck with was a conceptual one being: "How do we send tasks to the child processes we've made". As you can see we, thankfully, managed to figure out the need for a "if (cons_pid == 0)". This allowed us to prioritize child processes. In the next few lines we proceeded to have our program wait for our threads to finish which let us finally run our code deterministically.

```
[h703225011@adams20416 csc015_labs]$ ./gilab evee charizard salamace axew yveltal en
Vowel: evee
Vowel: axew
Vowel: en
Consonant: charizard
Consonant: salamace
Consonant: yveltal
[h703225011@adams20416 csc015_labs]$ ./gilab apple banana orange grape kiwi umbrella
Vowel: apple
Vowel: orange
Vowel: umbrella
Consonant: banana
Consonant: grape
Consonant: kiwi
[h703225011@adams20416 csc015_labs]$
```